Run-Chao Chen

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Research Profile

I study magnetar-powered astrophysical transients, including gamma-ray bursts (GRBs), magnetar giant flares (MGFs), fast X-ray transients, fast radio bursts, and superluminous supernovae. My recent studies focus on searching for (quasi-)periodicities in GRB prompt emission through timing analysis, seeking evidence for millisecond magnetars or, alternatively, an extragalactic MGF origin for some short GRBs. Building on these studies, my current interests lie in exploring how these rare coherent signals arise and under what conditions central-engine dynamics can leave observable signatures in the emission—or whether such signatures can be ruled out in specific scenarios.

Research interests: magnetars; time-domain astrophysics; multimessenger astronomy.

Education

Ph.D. in Astrophysics | Nanjing University, China 2023–2027 (expected)

School of Astronomy and Space Science
Dissertation: The physical origin and radiation mechanism of astrophysical transient powered by magnetar

M.S. in Physics | Guangxi University, China 2020–2023

School of Physical Science and Technology
Thesis: Research on magnetar bursts and their association with fast radio bursts

Research Experience

Member | NJU GRB Group 2023–present

My research in the NJU GRB Group focuses on the timing and spectral properties of GRBs. I developed a systematic framework to search for high-frequency quasi-periodic oscillations in bright GRBs and evaluate their statistical significance (Chen et al., 2025, Nature Astronomy; 2026, ApJ). I also performed temporal and spectral analyses of GRBs associated with Einstein Probe fast X-ray transients (Yin et al., 2024, 2025, ApJL), and assisted with undergraduate projects and theses on machine-learning methods for GRB identification and classification (2024–2026).

Member | GXU Observation Team 2020–2023

During my master's studies, I analyzed Fermi/GBM observations of soft gamma repeater bursts, focusing on their temporal and spectral properties and possible connections to fast radio bursts, motivated by FRB 200428 / SGR 1935+2154 (Chen et al., 2023, RAA). Alongside this work, I participated in optical observing shifts and proposal preparation for rapid GRB follow-up, long-term active galactic nucleus monitoring, and monitoring of repeating FRB sources using the LCOGT network and GWAC F-60 telescopes at Xinglong Observing Station.

Mission Roles and Observing

Associate Scientist and Burst Advocate | The SVOM Mission 2024–present

- Validate and assess ECLAIRs triggers, perform preliminary analyses, and disseminate event information through GCN Notices and Circulars. Typical duties include two days of follow-up support and three days of real-time ECL trigger validation per month.
- Contributed to SVOM science analyses, including the study of GRB 241030A (Varun et al., 2026, ApJ).

Associate Member and Transient Advocate | The Einstein Probe Mission 2024–present

- Contributed to studies of Einstein Probe transients, including EP240219a and EP250404a, as a member of Prof. Binbin Zhang's team.
- Since September 2026, undertake transient-advocate shifts averaging two to three days per month, including WXT trigger validation, observation instruction entry, and follow-up of identified fast X-ray transients.

Principal Investigator | Parkes (Murriyang) Proposal P1380 2025

- Searching for Coherent Radio Bursts from the Magnetar Remnant of GRB 230307A. Proposal P1380, semester 2025OCTS. Full allocation awarded; observations completed.
  Example dataset DOI: 10.25919/mgb2-ef72

Honors and Awards

2026 | National Scholarship for Graduate Students, Nanjing University
2025 | Graduate Talent Scholarship, Tier One, Nanjing University

Talks and Presentations

2026, Online | Tsinghua University "The GRID Project" Summer Camp — Invited talk

2026, Nanning, China | 10th Symposium of the Guangxi Key Laboratory for Relativistic Astrophysics — Oral presentation

2025, Beijing, China | Workshop on Extragalactic Fast Transients and Multimessenger Time-domain Astronomy — Oral presentation

2025, Yancheng, China | Conference on Compact Star Physics — Oral presentation

Publications

First-author papers

2026

Harmonic brightness oscillations point to magnetar giant flares in rare short gamma-ray bursts
R.-C. Chen, O. Gottlieb, B.-B. Zhang, et al., Science Advances,
https://www.science.org/doi/10.1126/sciadv.aee6388

A 4.5 s Quasiperiodic Spectral Oscillation in GRB 230307A: Evidence for Free Precession of a Postmerger Magnetar?
R.-C. Chen, J. Yang, B.-B. Zhang, et al., The Astrophysical Journal Letters,
https://doi.org/10.3847/2041-8213/ae6c1b

Detection of a Millisecond Periodicity in BATSE Short Gamma-Ray Bursts.
R.-C. Chen, J. Yang, Y.-H. I. Yin, B.-B. Zhang, The Astrophysical Journal,
https://doi.org/10.3847/1538-4357/ae3a96

2025

Evidence for a brief appearance of gamma-ray periodicity after a compact star merger.
R.-C. Chen, B.-B. Zhang, C.-W. Wang, et al., Nature Astronomy,
https://doi.org/10.1038/s41550-025-02649-w

2023

Detection of Quasi-periodic Oscillations in SGR 150228213.
R.-C. Chen, C.-M. Deng, X.-G. Wang, et al., Research in Astronomy and Astrophysics,
https://doi.org/10.1088/1674-4527/acdabe

Co-authored papers

2026

Short Spike, Long Story: Episode-dependent Shifts of Long-duration Type-I Gamma-Ray Bursts on the E p,z–E iso Plane
Ya-Hui Jiang; **Run-Chao Chen**; Zhen-Yu Yan et al., The Astrophysical Journal,
https://doi.org/10.3847/1538-4357/aeac16

Fast-cooling Synchrotron Prompt Emission from Internal Shocks in GRB 241030A
Varun; Bin-Bin Zhang; Xiao-Hong Zhao et al., The Astrophysical Journal,
https://doi.org/10.3847/1538-4357/ae2eaf

2025

Spectral Hardening Reveals Afterglow Emergence in Long-duration Fast X-Ray Transients: A Case Study of GRB 250404A/EP250404a.
Y.-H. I. Yin, Y. Fang, B.-B. Zhang, et al., The Astrophysical Journal Letters,
https://doi.org/10.3847/2041-8213/adf552

2024

Triggering the Untriggered: The First Einstein Probe-Detected Gamma-Ray Burst GRB 240219A and Its Implications.
Y.-H. I. Yin, B.-B. Zhang, J. Yang, et al., The Astrophysical Journal Letters,
https://doi.org/10.3847/2041-8213/ad8652

Repeated Partial Disruptions in a White Dwarf–Neutron Star or White Dwarf–Black Hole Merger Modulate the Prompt Emission of Long-duration Merger-type GRBs.
J.-P. Chen, R.-F. Shen, W.-J. Tan, et al., The Astrophysical Journal Letters,
https://doi.org/10.3847/2041-8213/ad7737

A Comptonized Fireball Bubble Fits the Second Extragalactic Magnetar Giant Flare GRB 231115A.
Y.-H. I. Yin, Z. J. Zhang, J. Yang, et al., The Astrophysical Journal Letters,
https://doi.org/10.3847/2041-8213/ad2839